

Solution Architecture

SKILL WALLET — A SmartBridge Product

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Solution Architecture Diagram



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders the interactive graphical interface. Allows banking staff and underwriting officers to enter client profiles, financial parameters, and visualize probability scores alongside final approval/rejection outcomes.	HTML5, CSS3, Streamlit / Flask UI Templates
API Gateway	Serves as the central interface communication link. Handles incoming client web requests, orchestrates request traffic flow, performs form payload deserialization, and maps fields to backend logic endpoints safely.	Flask Request Router, Werkzeug WSGI Engine
Microservices / Core Logic	Performs real-time data transformations, missing attribute corrections, feature scaling, and loads pre-trained serialized model matrices to compute real-time predictive classifications.	Python, scikit-learn, Pandas, NumPy, XGBoost, Joblib
Database / Storage	Maintains static raw model datasets for pipeline evaluations, manages model checkpoint binaries, and writes analytical auditing application logs for tracking model behavior over time.	Local File System, CSV flat files, Joblib Binary Serialization